



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – 24CI3201

A.Y. 2026 - 2027

TITLE OF THE PROJECT		
S. No.	University ID	Name
1	2420030233	K. Hitheshini
2	2420030664	T. Sree Vaishnavi
3	2420030661	K. Bhargavi
4	2420030660	K. Sai Vaishnavi
Name of the Guide:		Dr. Archana Kalidindi

Abstract

Title: The Adaptive Healthcare Appointment Management Platform with AI-Powered Smart Scheduling

The Adaptive Healthcare Appointment Management Platform with AI-Powered Smart Scheduling is a web-based application designed to simplify and automate the process of booking hospital appointments and managing doctor schedules. The system eliminates the drawbacks of traditional appointment methods, such as long waiting times, manual paperwork, scheduling conflicts, and communication delays, by providing a centralized digital platform for patients, doctors, and administrators. Patients can register, log in, search for doctors by specialization, check real-time availability, and book or cancel appointments conveniently. Doctors can manage their schedules, update available time slots, and monitor patient appointments, while administrators oversee user management, doctor records, and appointment operations through a secure dashboard. The platform incorporates AI-powered smart scheduling to optimize appointment allocation, minimize conflicts, and improve resource utilization. Built using React.js, Spring Boot, Java, PostgreSQL, and the Gemini AI API, the system offers a secure, responsive, and scalable solution. It enhances operational efficiency, improves communication between patients and healthcare providers, reduces waiting times, and delivers a seamless digital healthcare experience while supporting modern software engineering and cloud-based deployment practices.

Signature of the Guide

HOD